25900118 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR EXAMINATION MARCH 2025

SECOND SEMESTER

Discipline Specific Core Course (DSC) - MG2DSCIAM101 - AUTOMOTIVE SYSTEMS FOR E - VEHICLES

(2024 ADMISSION ONWARDS)

Duration: 1.5 Hours **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Multiple Choice Questions

Answer all questions

Each question carries **1** marks

1. A Unity Gain Amplifier using an op-amp is also known as: [K] / [CO1]

   a). Inverting amplifier b). Non-inverting amplifier

   c). Voltage follower d). Differential amplifier

2. What is commonly added to the output of a bridge rectifier to smooth the DC signal? [K] / [CO2]

   a). A voltage regulator b). A capacitor

   c). A transformer d). A resistor

3. What does BLDC stand for? [U] / [CO3]

   a). Bipolar Linear Direct Current b). Brushless Linear Direct Current

   c). Brushless DC d). Bi-directional Linear DC

4. In a photodiode amplifier circuit, the operational amplifier configuration is [U] / [CO1]

   a). Inverting amplifier b). Differential amplifier

c). Transimpedance amplifier d). Voltage comparator

5. Which sensor is commonly used in ESCs to monitor the motor's current and voltage in real time? [U] / [CO3]

a). Hall effect sensor b). Temperature sensor

c). Current and voltage sensor d). GPS sensor

6. In an inverting amplifier, which of the following resistors determines the gain of the amplifier? [K] / [CO1]

a). Input resistor (Rin) b). Feedback resistor (Rf)

c). Both Rin and Rf d). None of the above

7. In a comparator circuit using the 741 op-amp, if the reference voltage (inverting input) is set to 0V, what would be the output state when the non-inverting input exceeds 0V? [K] / [CO2]

a). Negative saturation b). Positive saturation

c). Zero volts d). It would oscillate

8. What is the primary function of an Electronic Speed Controller (ESC) in an Electric Vehicle (EV)? [U] / [CO3]

a). To control the power supply to the motor b). To manage the temperature of the battery

c). To regulate the speed of the motor d). None of these

9. What is the primary function of a 78XX voltage regulator IC? [U] / [CO2]

a). To increase voltage

b). To provide a constant output voltage

c). To store electrical energy

d). To convert AC to DC

10. The rotor current in an induction motor is induced by: [U] / [CO3]

a). External power supply b). The magnetic field produced by the stator

c). A DC generator d). A rotating mechanical force

[1x10 = 10]

Part B

Short Answer Questions

Answer any **4** questions

Each question carries **5** marks

11.	Compare single, dual and quad packages.	[U] / [CO1]
12.	Briefly explain the working of Schmitt Trigger.	[U] / [CO2]
13.	Explain the basic structure of a lithium-ion battery.	[K] / [CO2]
14.	Describe the symbol of an operational amplifier, its components, and explain the significance of each part in the circuit.	[U] / [CO1]
15.	What role does the switching transistor play in an SMPS?	[U] / [CO2]
16.	What is DC fast charging, and why is it important?	[U] / [CO3]

[5x4 = 20]

Part C

Essay Questions

Answer any **2** questions

Each question carries **10** marks

17.	Explain open loop op amp configurations.	[U] / [CO1]
18.	Define a Battery Management Unit (BMU). Describe its primary functions, components, and their roles in ensuring safe and efficient battery operation.	[U] / [CO3]
19.	Discuss the effect of a capacitor filter in a bridge rectifier circuit.	[U] / [CO2]
20.	Describe the various types of EV charging connectors used globally. Discuss their specifications and compatibility with EVs.	[K] / [CO3]

[10x2 = 20]